

Final: Hopfield Network

Group 3

VNUHCM - University of Science

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23122001 - Dinh Duc Anh Khoa
23122006 - Luu Thuong Hong
23122008 - Mai Duc Minh Huy
23122020 - Nguyen Thien An

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1.1. Introduction

Associative Memory

- Retrieves patterns from partial or noisy input data, a foundational concept in machine learning.

Core Characteristics

- A type of attractor network.
- Facilitates auto-association (pattern completion) and optimization tasks in neural networks.
- Operates as a Recurrent Neural Network (RNN) leveraging the Hebbian learning rule [Hertz et al., 1991].

Limitations of Traditional RNNs

- Face performance bottlenecks and storage inefficiencies, addressed by the Hopfield Network.



1.2. Why Hopfield Network?

- Recognized by the Nobel Prize in Physics 2024.
- Integrates self-attention mechanisms in Transformers [Vaswani et al., 2017] [Ramsauer et al., 2020].

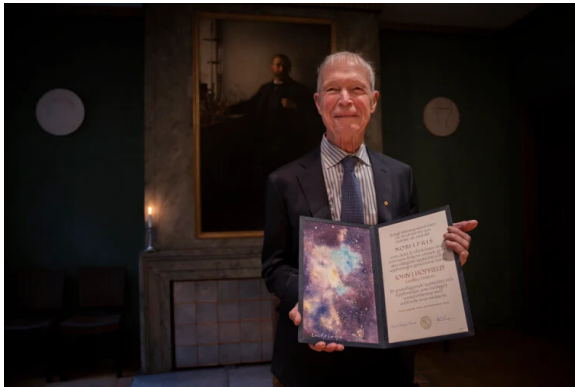
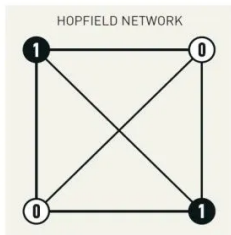


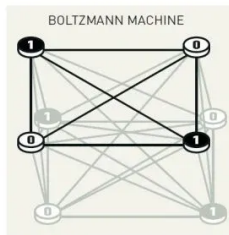
Photo: Nanaka Adachi

2. Hopfield Network and the concepts behind it

Different types of network

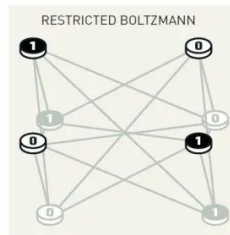


John Hopfield's associative memory is built so that all the nodes are connected to each other. Information is fed in and read out from all the nodes.



Visible nodes Hidden nodes

Geoffrey Hinton's Boltzmann machine is often constructed in two layers, where information is fed in and read out using a layer of *visible* nodes. They are connected to *hidden* nodes, which affect how the network functions in its entirety.

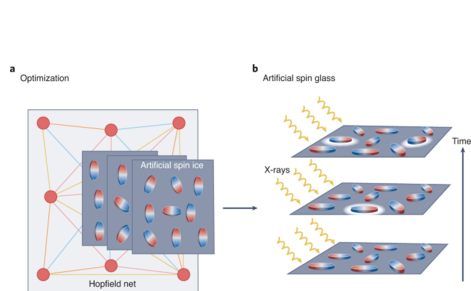


In a restricted Boltzmann machine, there are no connections between nodes in the same layer. The machines are frequently used in a chain, one after the other. After training the first restricted Boltzmann machine, the content of the hidden nodes is used to train the next machine, and so on.

Figure 1: Different type of network [Dave, 2024]

2. Hopfield Network and the concepts behind it

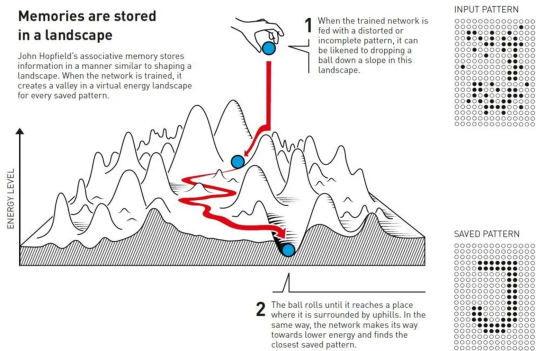
The inspiration for Hopfield network comes from spin-glass systems in physics [Müller et al., 1995]. Hebbian Learning shapes the energy landscape by creating deep wells at stored pattern locations. The network then transitions between states, with noisy inputs naturally rolling down to the nearest stored pattern.



Spin-glass system [Makarov, 2022]

Memories are stored in a landscape

John Hopfield's associative memory stores information in a manner similar to shaping a landscape. When the network is trained, it creates a valley in a virtual energy landscape for every saved pattern.



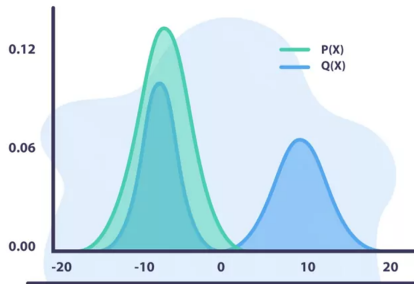
Energy Landscape [Dave, 2024]

2.1 FEP [Parr et al., 2022]

- Free energy is the difference between a system's predictions of upcoming inputs and the incoming inputs. (*Equivalent to loss function*)
- The free-energy principle states that biological systems (a neuron, the brain, ...) will try to minimize free energy by **updating its model of the world toward better predictions** (KL-Divergence) OR **acting to change the world to agree with its model** (Action-based Inference).

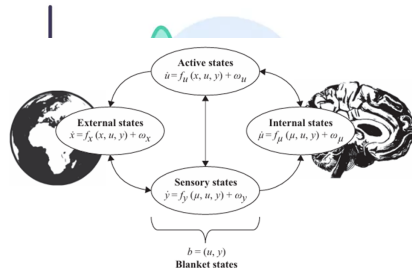


2.1 FEP [Parr et al., 2022]



Kullback–Leibler Divergence

$$D_{\text{KL}}(p(x)||q(x)) = \int p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx$$



Kullback–Leibler Divergence

Markov Blanket States

2.2 Hebbian Learning [Albanese et al., 2024]

- Hebbian learning is a learning mechanism inspired by the brain's plasticity, where the connection between neurons is strengthened if they are activated simultaneously.
- Main idea: “*Neurons that fire together, wire together*”.
→ Neurons that fire out of sync lose their link.

The basic premise of Hebbian Learning is encapsulated in the formula:

$$\Delta w = \eta xy$$

where:

- Δw is the change in the weight of the synapse,
- η is the learning rate, a small positive scalar that determines the step size of the weight update,
- x is the presynaptic neuron's activity level, and
- y is the postsynaptic neuron's activity level.



3.1. Classic Hopfield Network [Hopfield, 2007]

- Stored patterns: $\{x_i\}_{i=1}^N$, where:
 - N : number of stored patterns.
 - x_i must be a polar pattern with d elements.
- Weight matrix:

$$W = \sum_i^N x_i x_i^T$$

- Main goal: Retrieving the original pattern from a noisified one.



3.1.1 Synchronous

Update rule: Update pattern S according to the following formula:

$$S^{t+1} = \text{sgn}(WS^t - b),$$

where:

- $\text{sgn}(\cdot)$: sign function.
- b : bias vector (typically $\vec{0}$).
- S^t : state pattern at the t -th update.

Convergence occurs when $S^{t+1} = S^t$.



3.1.2 Asynchronous

Energy function:

$$E(S) = -\frac{1}{2}S^T W S + S^T b$$

Update rule: Update the components s_i of the vector S according to the formula:

$$s_i = \text{sgn}(W s_i - b),$$

where:

- s_i : the i -th component of the pattern S .

Convergence is achieved when S is a local minima of the function $E(S)$.



3.1. Classic Hopfield Network [Hopfield, 2007]

d is the dimension of the input.

Limitations

- Small memory capacity: $C \approx 0.14d$.
- Not reliable enough in retrieving original patterns.
- Only support polar (binary) patterns.



3.2. Modern Hopfield Network

Krotov and Hopfield introduced the energy function:

$$E = - \sum_{x=1}^N F(x_i^T S),$$

where $F(z) = z^\alpha$ is a polynomial interaction function and N is the number of stored patterns.

New update rule:

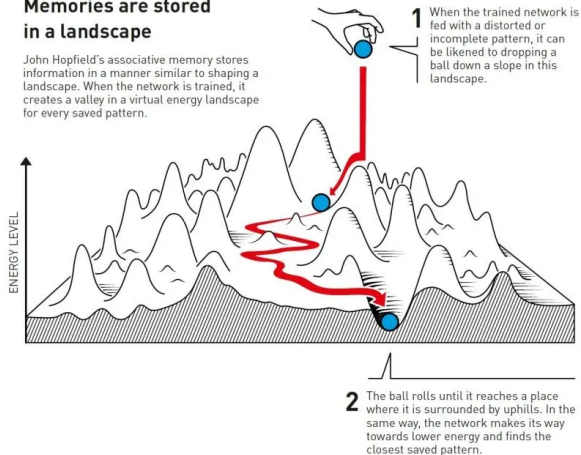
$$S^{\text{new}}[l] = \text{sgn} \left[-E(S^{(l+)}) + E(S^{(l-)}) \right],$$



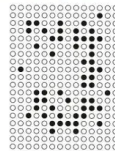
3.2. Modern Hopfield Network

Memories are stored in a landscape

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INPUT PATTERN



SAVED PATTERN

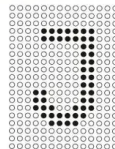


Figure 2: Pattern approaching local minima

3.2. Modern Hopfield Network

Pros:

- Memory capacity is much higher: $C = \frac{1}{2(2\alpha - 3)!!} \frac{d^{\alpha-1}}{\log(d)}$.
- Achieves convergence after one update.

Cons:

- Might converge to incorrect local minima.



4. Application

- Associative Memory for Image Reconstruction (Simpson Dataset).

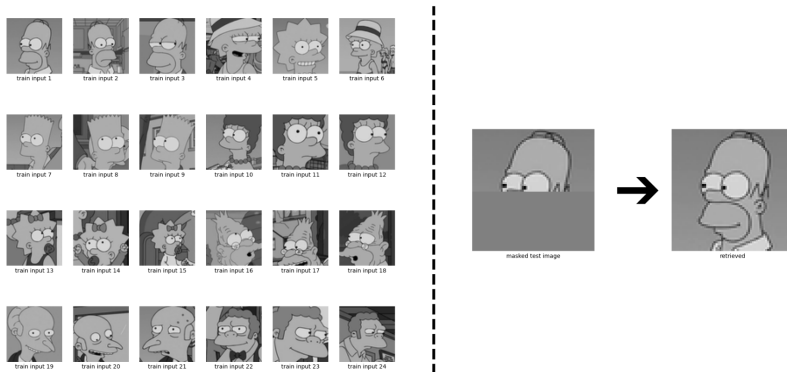


Figure 3: Image Reconstruction

4. Application

- Hopfield Attention [Vaswani et al., 2017].

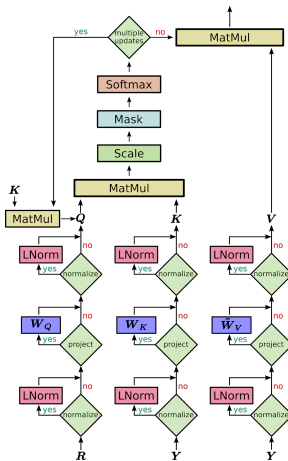


Figure 4: Hopfield Attention

4. Application

- NP-hard solver (N-queens problem) [Mańdziuk, 1995].

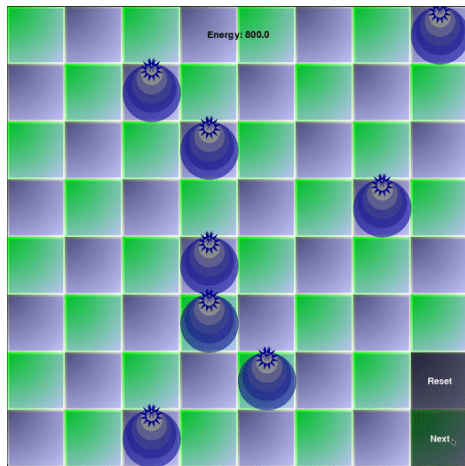
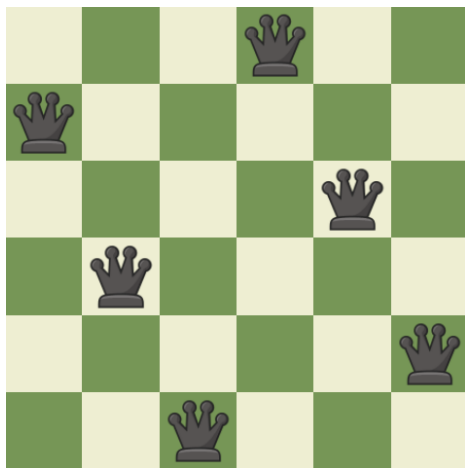


Figure 5: N-queens problem

4.3 N-queens problem solver

The N-queens problem is a classic combinatorial problem where the goal is to place n chess queens on an $n \times n$ chessboard such that no two queens threaten each other.



4.3 N-queens problem solver

The Hopfield network model used is the classical binary Hopfield model, modified to be suitable for solving the N-Queens problem.

Compared to the original model, this model has the following main differences:

- Completely binary.
- Random initialization.
- Modified energy function and input potential computation.

Goal: Find a state configuration that minimizes the energy function, or ideally, makes the energy function equal to zero.



4.3 N-queens problem solver

Energy function

The **energy function** E is defined as follows:

$$E = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n [A \cdot R_{ij} + B \cdot D_{ij}] + \frac{1}{2} C \left(\sum_{i=1}^n \sum_{j=1}^n v_{ij} - (n + \sigma) \right)^2$$

Where:

- $R_{ij} = \left(\sum_{k \neq j} v_{ik} + \sum_{k \neq i} v_{kj} \right) v_{ij}$ (penalizes when there is more than one Queen in the same row or column)
- $D_{ij} = \left(\sum_{(k,l) \in \text{diagonal of } (i,j)} v_{kl} \right) v_{ij}$ (penalizes if there is another Queen on the same diagonal)
- $\sum v_{ij}$: Total number of Queens on the board



4.3 N-queens problem solver

Output potential function

Binary threshold function to compute the **output potential** of the neuron:

$$v(u) = \begin{cases} 0 & \text{if } u < 0 \\ 1 & \text{if } u \geq 0 \end{cases}$$



4.3 N-queens problem solver

Input potential function

Given the **input potential** of the neuron (i, j) at time t , the formula for the input potential $U_{ij}(t)$ is:

$$U_{ij}(t) = -A \cdot H_{ij}(t) - R_{ij}(t) - S_{ij}(t) - C \cdot \left(\sum V_{kl}(t) - (n + \sigma) \right) \quad (1)$$

- $H_{ij}(t) = \sum_{k \neq j} V_{ik}(t) + \sum_{k \neq i} v_{kj}(t)$: Sum of states over rows and columns
- $R_{ij}(t)$: Off-diagonal interaction force (when $i - j = \text{const}$)
- $S_{ij}(t)$: Sum of on-diagonal interaction forces (when $i + j = \text{const}$)
- $\sum v_{kl}(t)$: Total number of queens on the board at time t



4.3 N-queens problem solver

Experimentation

Due to limitations in time and computational resources, we conducted experiments with 4, 8, and 16 queens. For each configuration, we performed 100 runs and calculated the convergence rate of that configuration.

We will employ three network initialization strategies:

- **Strategy 1:** All neurons set to 0.
- **Strategy 2:** Neurons randomly set to 0 or 1.
- **Strategy 3:** All neurons set to 1.



4.3 N-queens problem solver

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4.3 N-queens problem solver

Experimentation

	Strategy 1	Strategy 2	Strategy 3
$n = 4$	100%	100%	100%
$n = 8$	100%	99%	100%
$n = 16$	100%	100%	100%

Table 1: $A = B = C = 100$ and $\sigma = 0$ (asynchronous mode)

	Strategy 1	Strategy 2	Strategy 3
$n = 4$	100%	100%	100%
$n = 8$	98%	100%	97%
$n = 16$	100%	100%	99%

Table 2: $A = B = 100$, $C = 40$ and $\sigma = 2$ (synchronous mode)



4.3 N-queens problem solver

Experimentation

	Strategy 1	Strategy 2	Strategy 3
$n = 4$	100%	100%	100%
$n = 8$	100%	100%	100%
$n = 16$	100%	100%	100%

Table 3: $A = 100$, $B = 50$, $C = 100$ and $\sigma = 0$ (asynchronous mode)

	Strategy 1	Strategy 2	Strategy 3
$n = 4$	18%	17%	27%
$n = 8$	4%	5%	7%
$n = 16$	3%	1%	2%

Table 4: $A = B = 100$, $C = 40$ and $\sigma = 0$ (synchronous mode)



4.3 N-queens problem solver

Time complexity

Calculating the complexity for this problem is very challenging. In the original paper, the author estimated the complexity experimentally with the following results:

- Asynchronous mode: $O(n^{4.47})$
- Synchronous mode: $O(n^{4.67})$



4.3 N-queens problem solver

Advantages and disadvantages

- **Advantages:**

- High convergence rate
- Effective for NP-Hard problems
- Stability

- **Disadvantages:**

- Hyperparameter tuning required
- Complex energy function



4.4 Wordle

- Wordle.

Wordle

E	N	J	O	Y
O	T	H	E	R
W	O	R	D	Y
G	A	M	E	S
T	O	D	A	Y

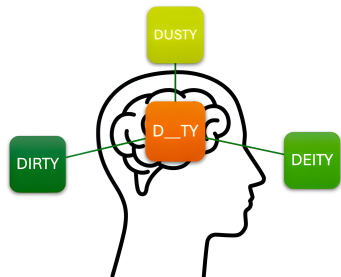
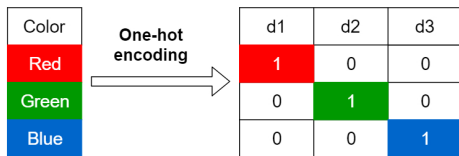


Figure 7: Wordle

4.4 Wordle

Methodology

- Wordle.



$$\begin{matrix} \boxed{M} \\ m \times n \end{matrix} \approx \begin{matrix} \boxed{L_k} \\ m \times k \end{matrix} \times \begin{matrix} \boxed{R_k^T} \\ k \times n \end{matrix}$$

Figure 8: Method

4.4 Wordle

Result & Ideas

Result: Failed

- Sub-optimal energy minima: Cause the network to converge into nonsensible words.
- Suppress mechanism for newly recalled words: Delete permanently from memories.
 $\implies$ Unstable, cause havoc and chaos.

Ideas:

- Inhibitory field in Hopfield.
- Dense distributed representation. Help me



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